

FedMosaic: Federated Retrieval-Augmented Generation via Parametric Adapters

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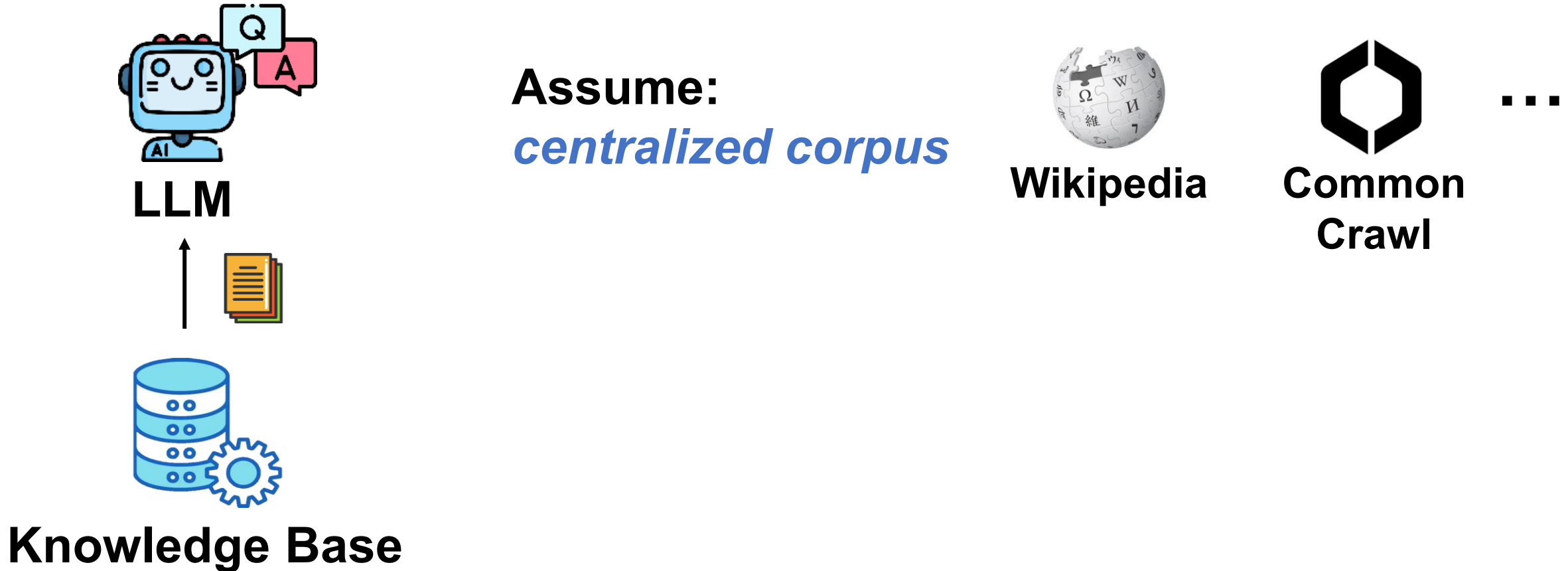


- **Background & Motivation**
- **Problem Statement**
- **Our Solutions**
- **Experiments**
- **Conclusion**

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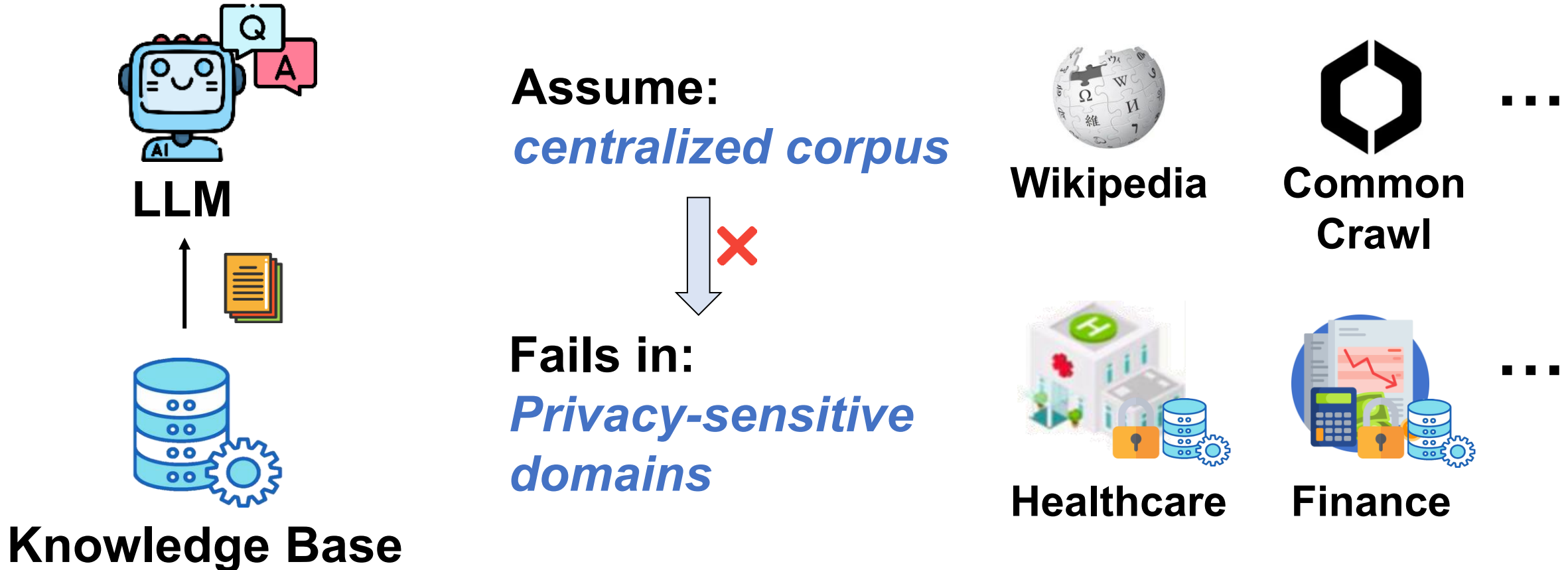
Background & Motivation

- **Retrieval Augmented Generation (RAG)**
 - Enhances LLMs with dynamically retrieved external knowledge



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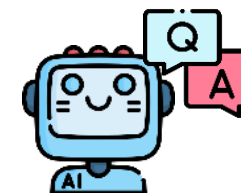
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Problem Statement

- **Federated Retrieval Augmented Generation (FedRAG)**
 - An LLM server with multiple silos



Server



LLM



Silos



Silo 1



Silo 2

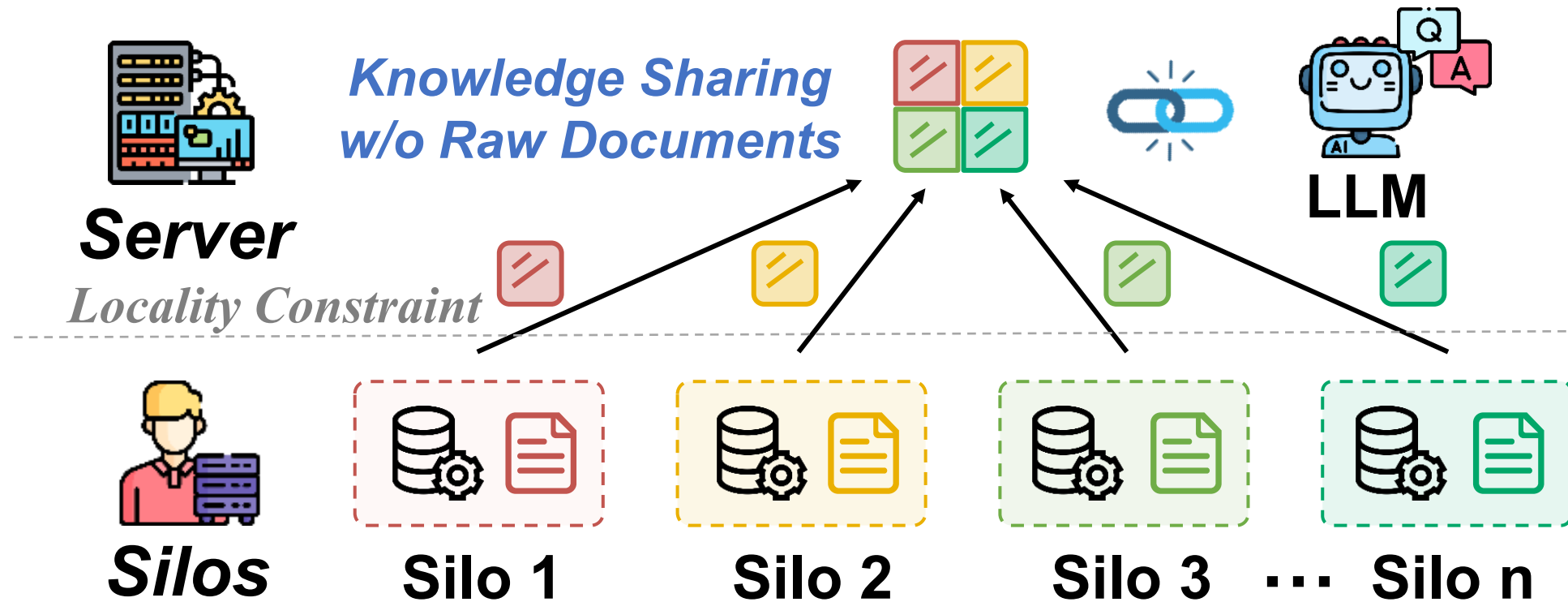


Silo 3 ... Silo n



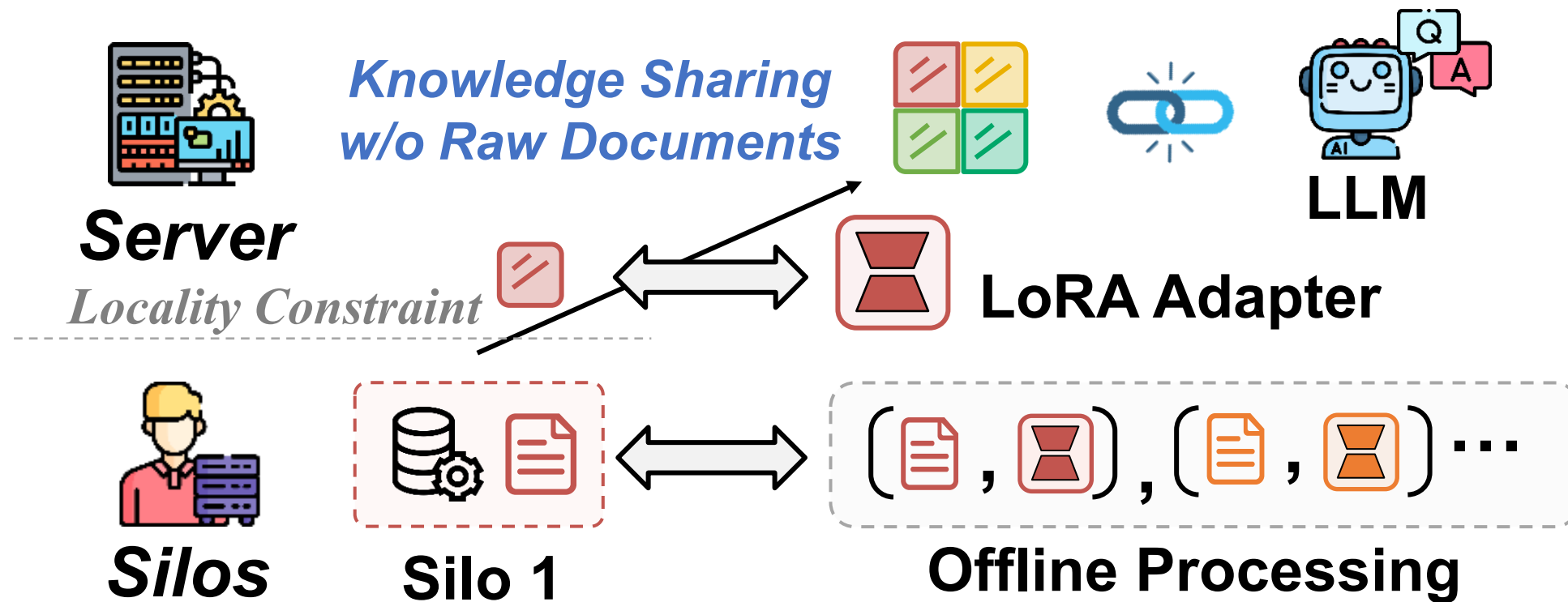
Problem Statement

- **Federated Retrieval Augmented Generation (FedRAG)**
 - An LLM server with multiple silos, and locality constraint



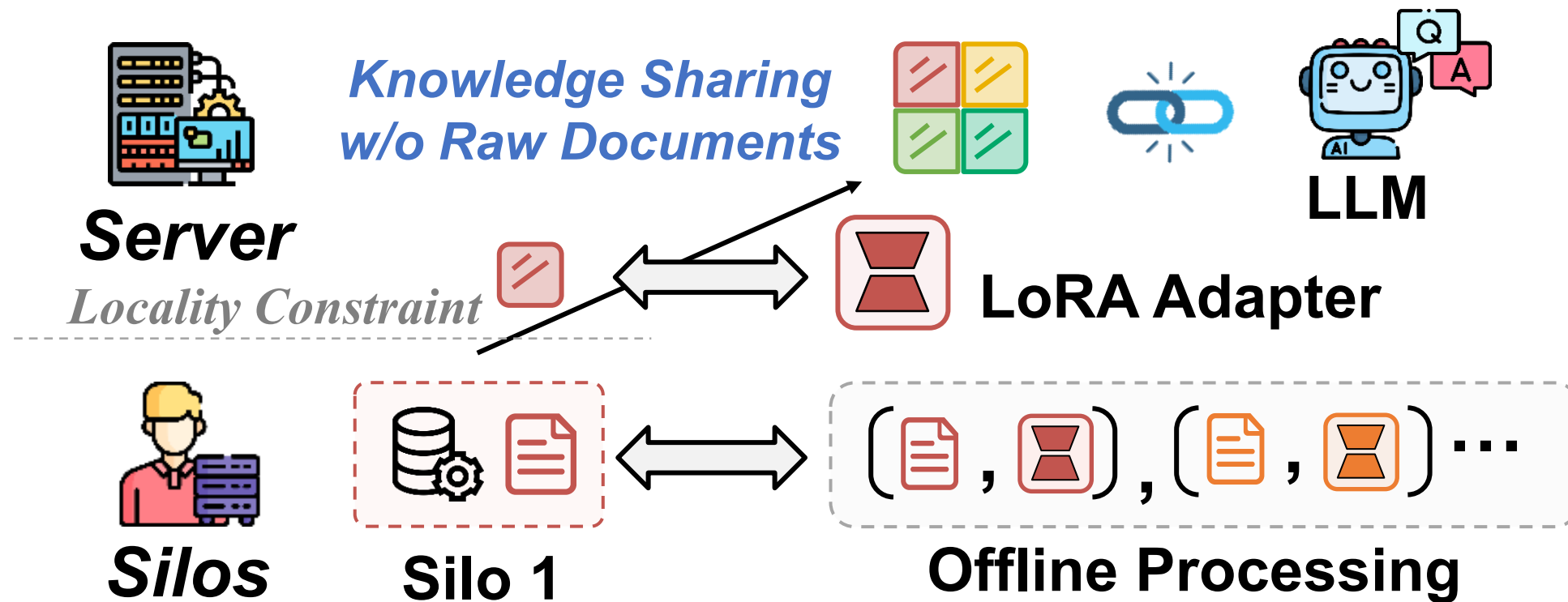
Problem Statement

- **Parametric Retrieval Augmented Generation (PRAG)**
 - Documents are encoded as **LoRA** adapters for inference



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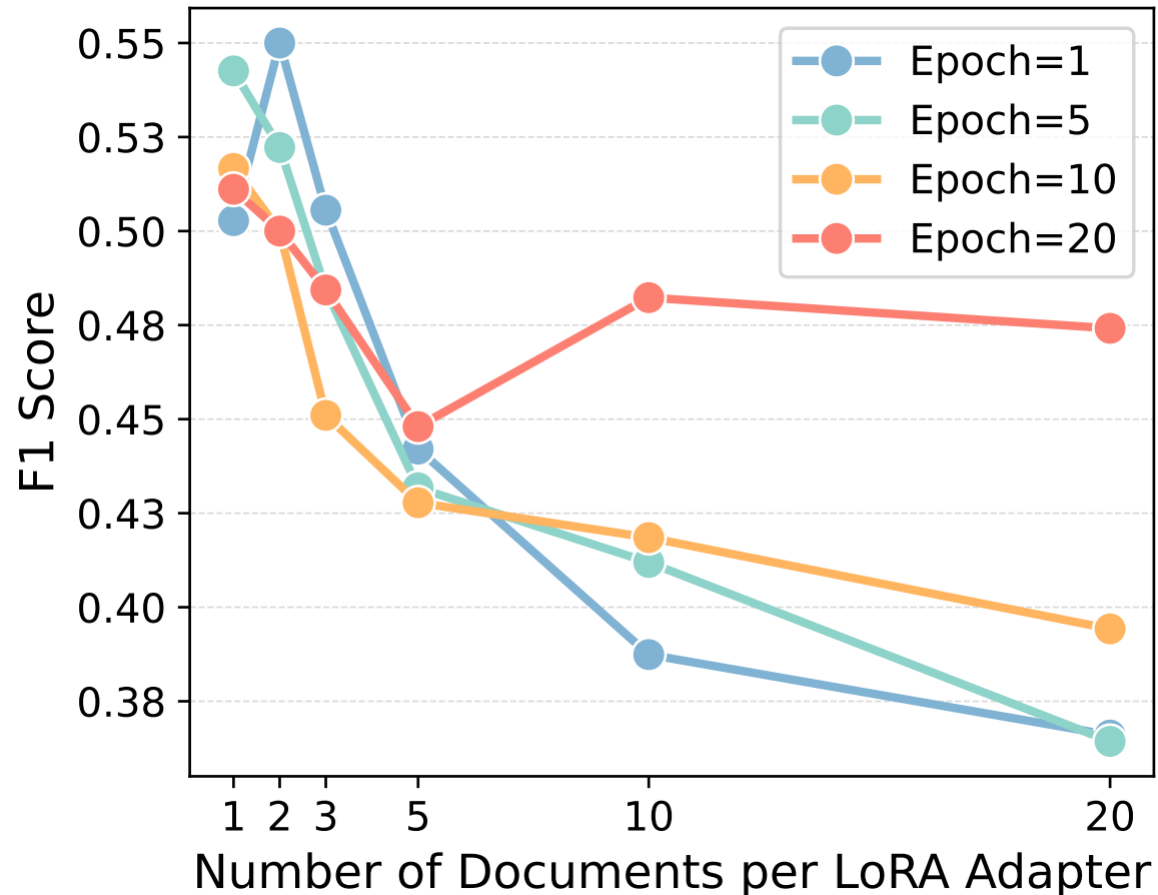


Key Idea: Parametrically sharing knowledge across silos

Problem Statement

● Challenge 1

● Storage and Communication Overhead



● one doc $\longleftrightarrow$ one LoRA

✗ *too high overhead*

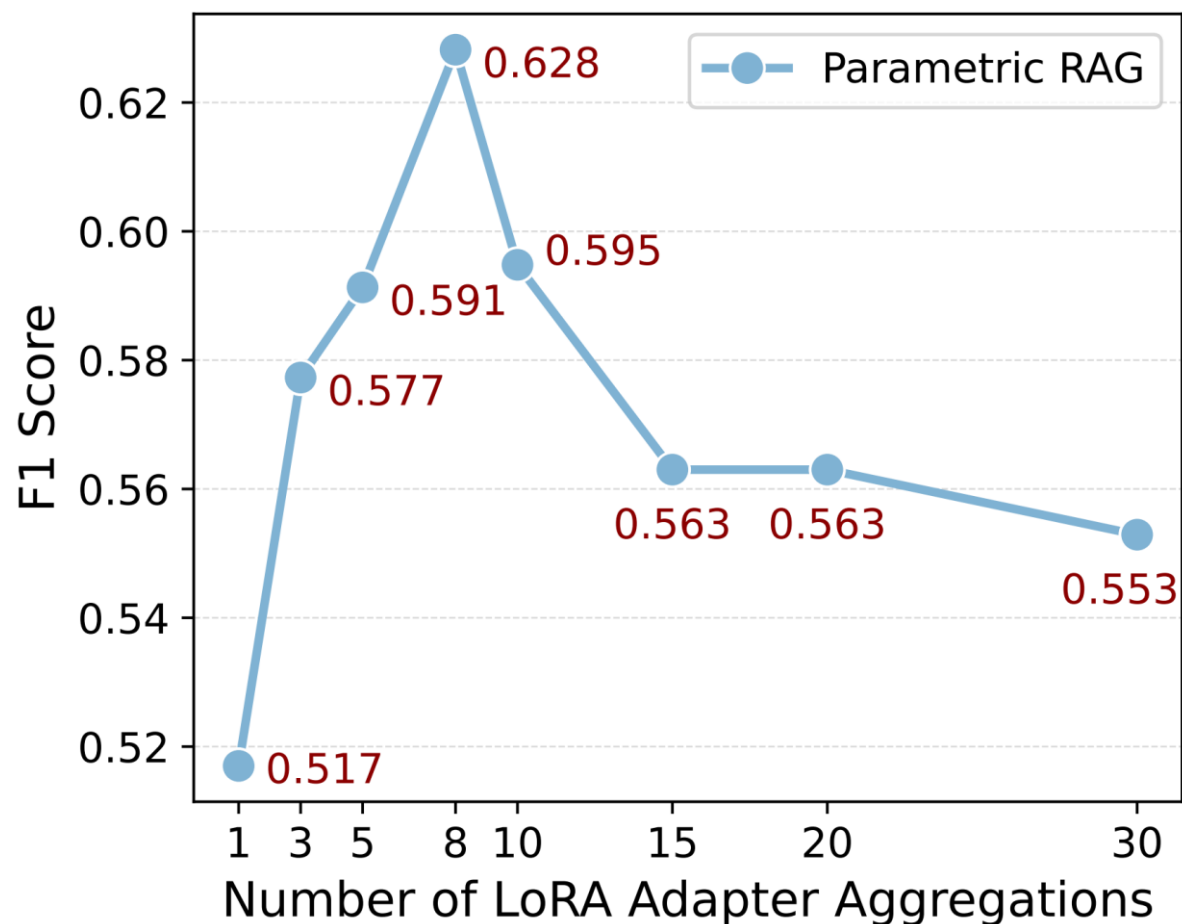
● clustered docs $\longleftrightarrow$ one LoRA

✗ *accuracy degradation*

Problem Statement

● Challenge 2

● Destructive Adapter Aggregation



● averaging all adapters



initially improves



then degrades accuracy



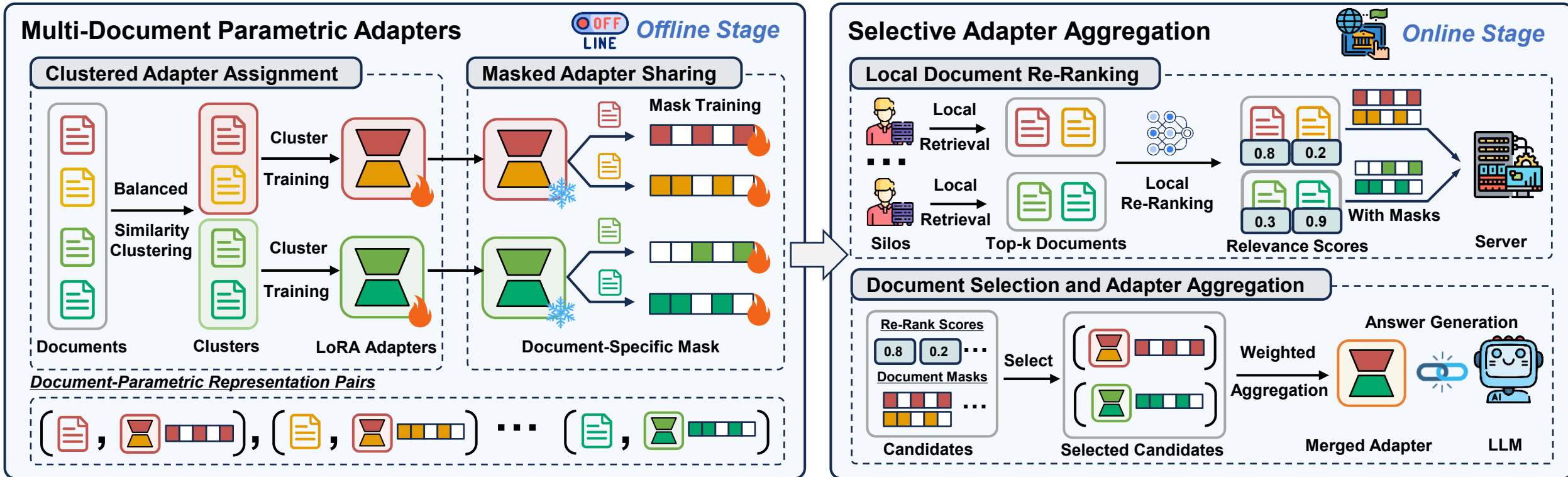
*parameter conflicts
among adapters*

Outline

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Our Solutions

● Overview of FedMosaic

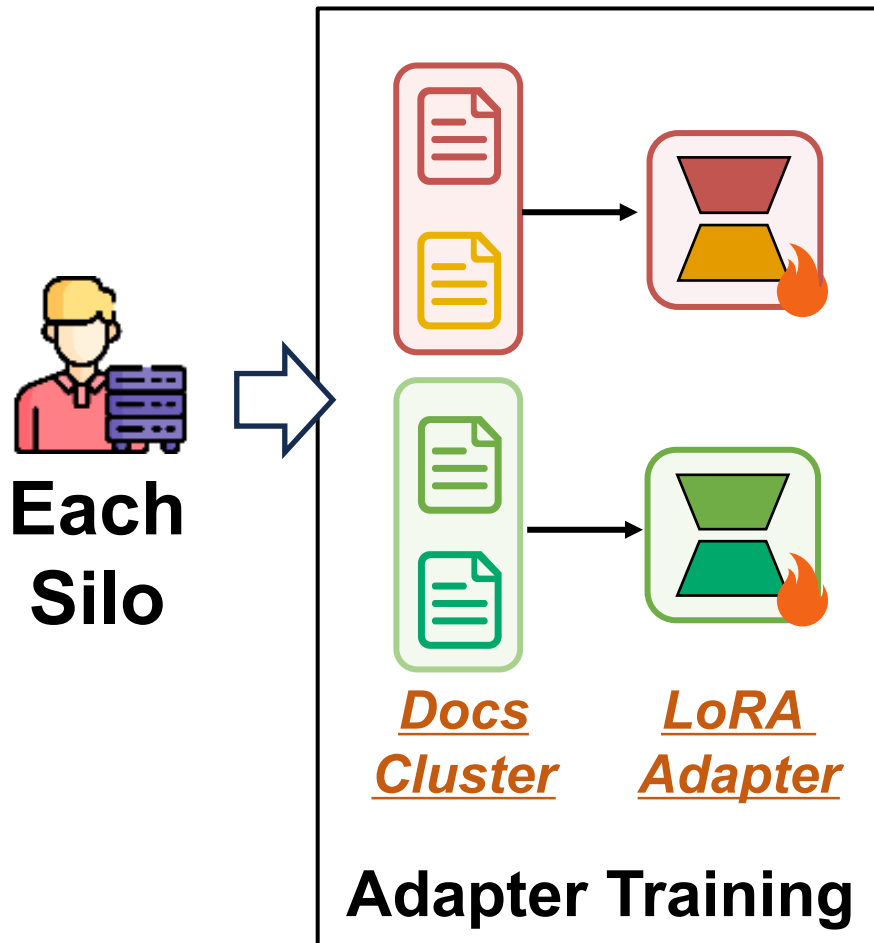


Workflow: ① Clustering – ② Mask Training – ③ Ranking – ④ Aggregating

Offline

Online

- **Offline Stage: Clustered Adapter Assignment**
 - **Balanced document clustering and cluster-level adapter training**



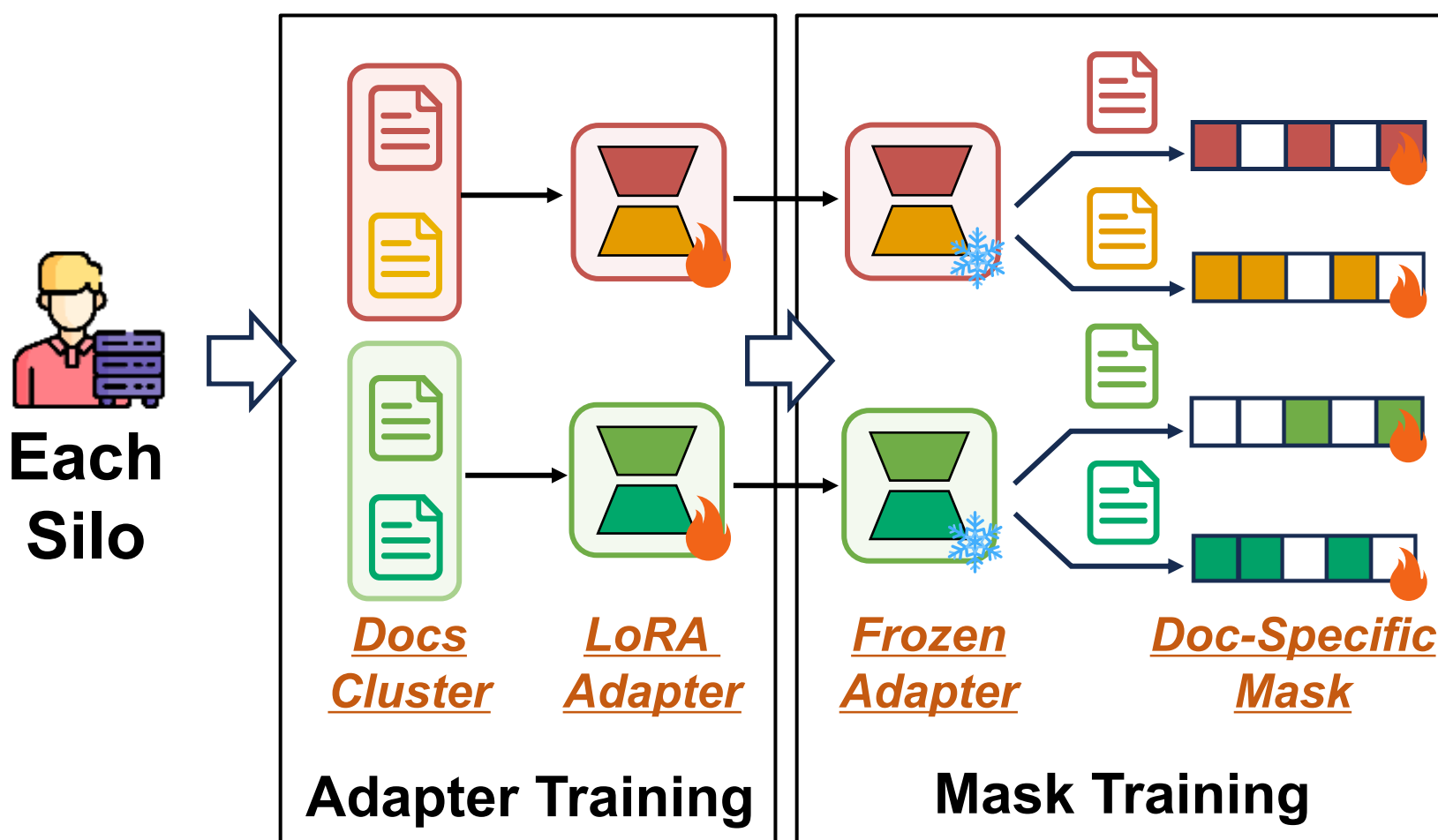
loss function

$$\min_{\Delta\theta} \sum_{x \in D^i} \sum_{t=1}^T -\log P_{\theta+\Delta\theta}(x_t \mid x_{<t}),$$

***Standard next-token loss
within each document cluster***

Our Solutions

- Offline Stage: *Masked Adapter Sharing*
 - Training a per-document mask with frozen cluster-adapter param.



loss function

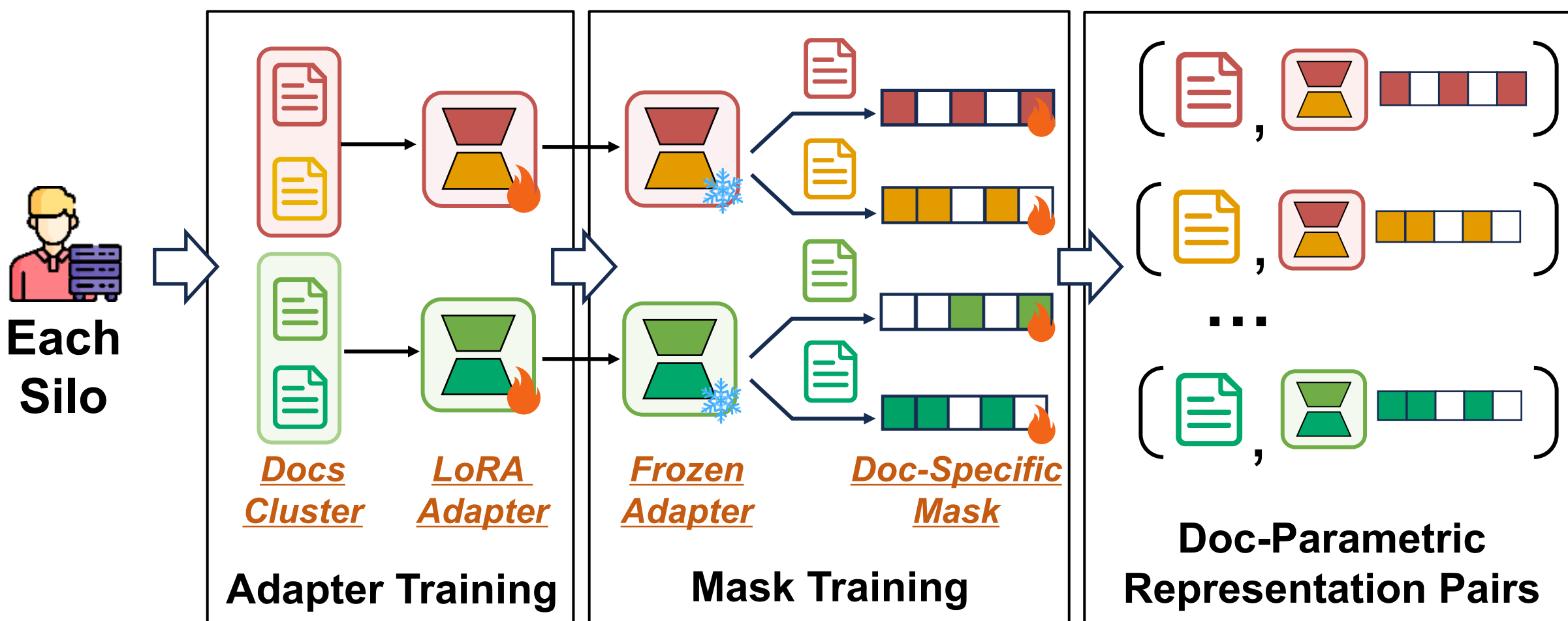
$$\min_{\hat{M}_m^{ij}} \mathcal{L}(\hat{M}_m^{ij}; \theta_F) = \mathcal{L}_{\text{next}}(\hat{M}_m^{ij}; \theta_F) + \lambda_{\ell_1} \left\| \sigma(\alpha \cdot \hat{M}_m^{ij}) \right\|_1$$

Next-token loss
+
Sparsity penalty

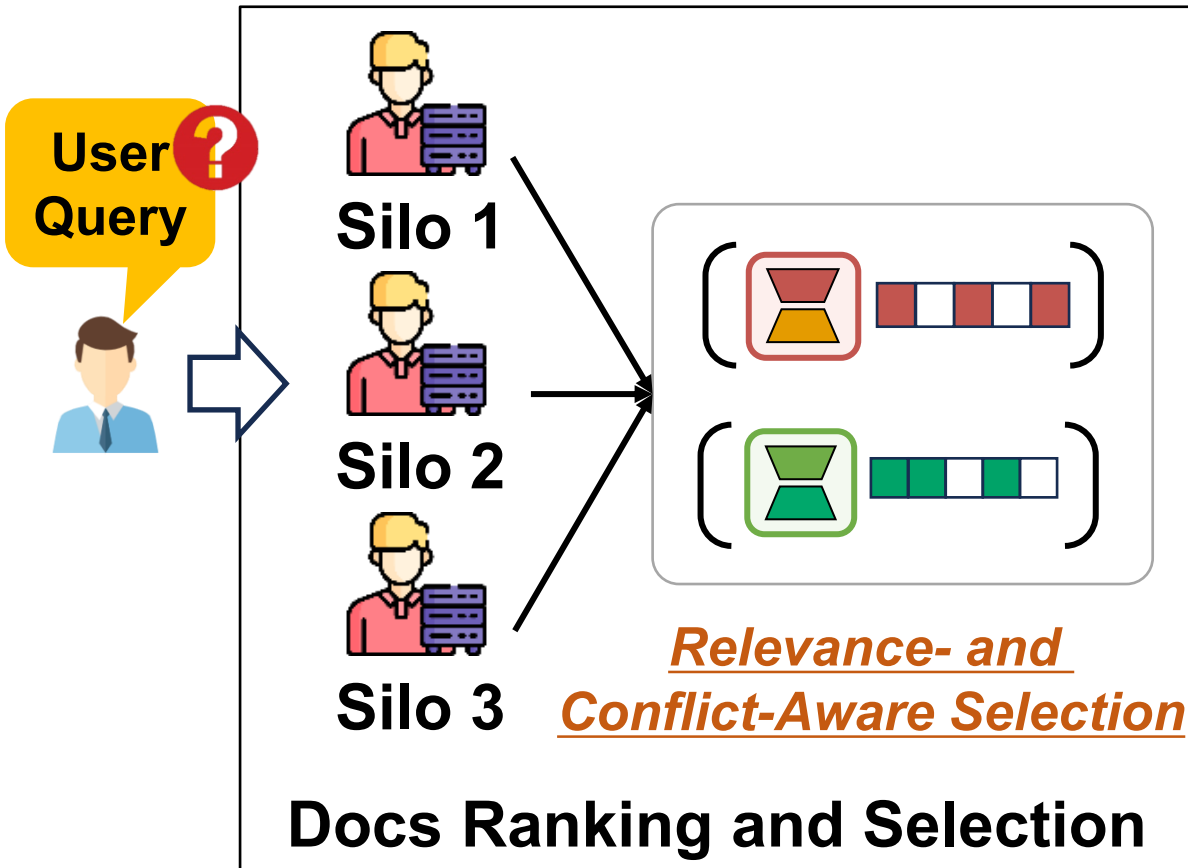
Our Solutions

- Offline Stage

- Each silo maintains a parameterized document repository



- Online Stage: *Document Ranking and Selection*
 - The server selects documents relevant to the user query



Objective:

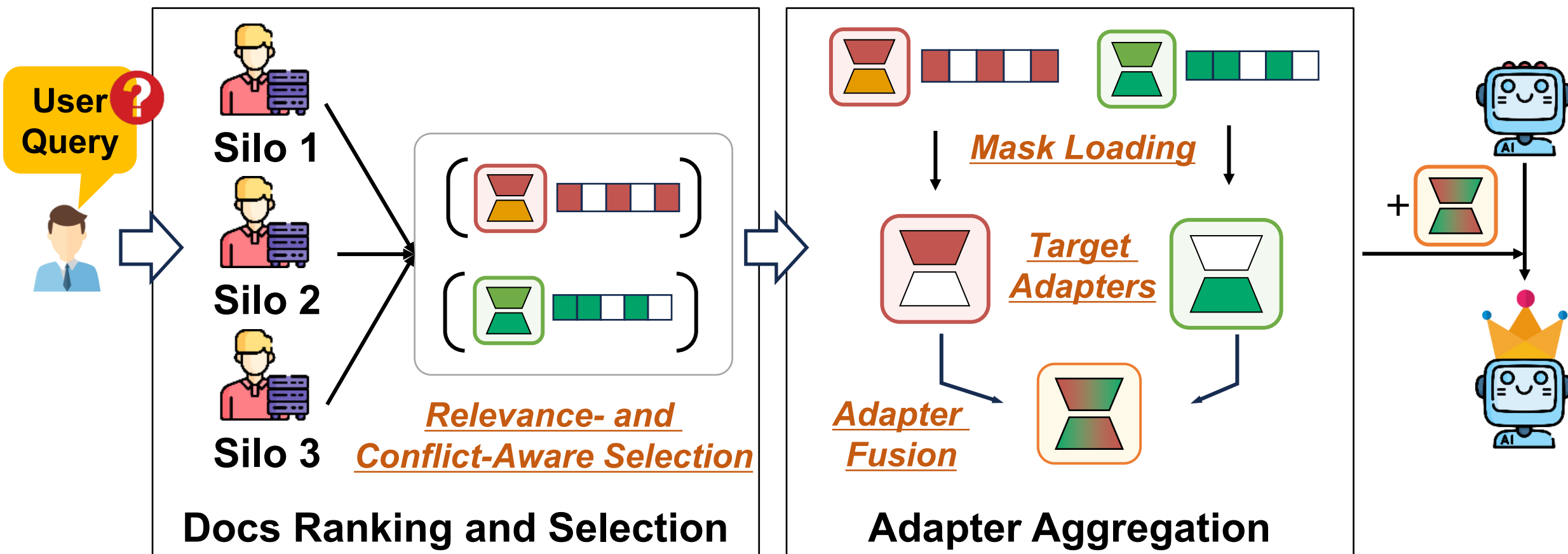
$$\max_{S \subseteq [n], |S|=k'} \left(\sum_{i \in S} \frac{s_i}{k'} - 2\lambda_{\text{ol}} \cdot \frac{\sum_{i,j \in S, i < j} \text{overlap}(M_i, M_j)}{k' \cdot (k' - 1)} \right)$$

where s_i denotes the relevance score

Relevance–Conflict Trade-off

Our Solutions

- **Online Stage: Masked Adapter Aggregation**
 - The adapter is combined with the base LLM to generate the answer



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● Accuracy

Method	HotpotQA		2WikiMultihopQA				PopQA	CWQ
	Bridge	Compare	Bridge	Compare	Inf.	Compose	Total	Total
Standard RAG	0.1474	0.3602	0.2838	0.2837	0.1549	0.0748	0.1366	0.2804
CoTRAG [38]	0.0808	0.2632	0.3821	0.3444	0.1380	0.0377	0.0444	0.2378
React [43]	<u>0.1731</u>	0.3103	0.3108	0.2608	0.1528	0.0590	0.1135	0.2443
Dargin [29]	0.1257	0.3540	0.3271	0.3786	0.1473	0.0766	0.0895	0.3662
FRAG [1, 49]	0.1317	0.3390	0.3357	0.2964	0.1527	0.0265	0.1207	0.1946
MKPQA [28]	0.1562	0.2757	0.2831	0.2679	0.2112	0.0510	0.1329	0.2008
RAGRoute [13]	0.1137	0.2825	0.2239	0.2472	0.1351	0.0293	0.0380	0.1767
FedIT [48]	0.1107	0.4188	<u>0.4250</u>	0.4377	0.1996	0.0692	<u>0.2152</u>	<u>0.3687</u>
FLoRA [37]	0.1030	<u>0.4259</u>	0.3625	0.4038	0.1864	<u>0.0770</u>	0.1917	0.3497
PRAG [30]	0.0983	0.4161	0.3875	<u>0.4805</u>	<u>0.2138</u>	0.0462	0.0759	0.2810
FedMosaic	0.1980	0.4547	0.4453	0.5275	0.2474	0.0940	0.2368	0.3841

FedMosaic achieves 10.9% higher accuracy on average

● Micro-Benchmarks

Attack	Method	Target Prompts	Target Info.	Repeat Prompts	Repeat Length
Target	IC-FedRAG	138.93	102.20	131.33	21.34
	FedMosaic	0	0	0	0
Prefix	IC-FedRAG	52,86	46.66	55.93	23.58
	FedMosaic	44.00	29.53	22.86	22.22

FedMosaic provides stronger resistance to data extraction attacks.

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Type	Method	2WikiMultiHopQA			
		Bridge	Compare	Inf.	Compose
Local	ReAct	0.3432	0.4022	0.2093	0.0746
FedRAG	MKPQA	0.2238	0.4011	<u>0.2797</u>	0.0932
FedFT	FedIT	<u>0.4995</u>	<u>0.6105</u>	0.2677	<u>0.1311</u>
FedFT	FLoRA	0.4701	0.5714	0.2355	0.1138
Param.	PRAG	0.1812	0.2527	0.0829	0.0271
Ours	FedMosaic	0.5207	0.6237	0.2859	0.1584

FedMosaic scales effectively to support substantially larger model sizes.

Experiments

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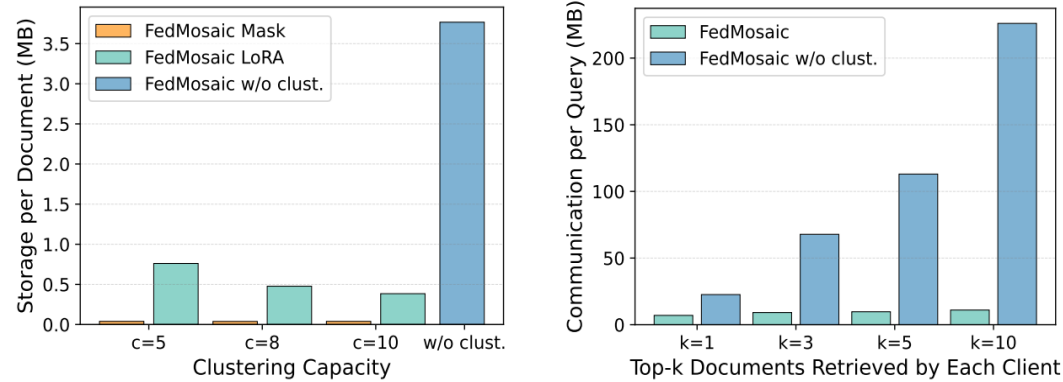
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Method	HotpotQA		2WikiMultihopQA				PopQA	CWQ
	Bridge	Compare	Bridge	Compare	Inf.	Compose	Total	Total
DP_Prompt	0.0228↓88.5%	0.2179↓52.1%	0.0785↓82.4%	0.1305↓75.3%	0.0385↓84.4%	0.0114↓87.9%	0.0060↓97.5%	0.0778↓79.7%
Sage (Attr)	0.0791↓60.1%	0.3286↓27.7%	0.1808↓59.4%	0.2218↓58.0%	0.1318↓46.7%	0.0222↓76.4%	0.0676↓71.5%	0.1217↓68.3%
Sage (Agent)	0.0923↓53.4%	0.3314↓27.1%	0.2898↓34.9%	0.2645↓49.9%	0.1343↓45.7%	0.0231↓75.4%	0.0560↓76.4%	0.1485↓61.3%
AUGPE	0.0079↓96.0%	0.0514↓88.7%	0.0153↓96.6%	0.0283↓94.6%	0.0225↓90.9%	0.0032↓96.6%	0.0004↓99.8%	0.0247↓93.6%

Privacy-preserving prompt engineering methods show severe accuracy degradation

● Ablation Studies

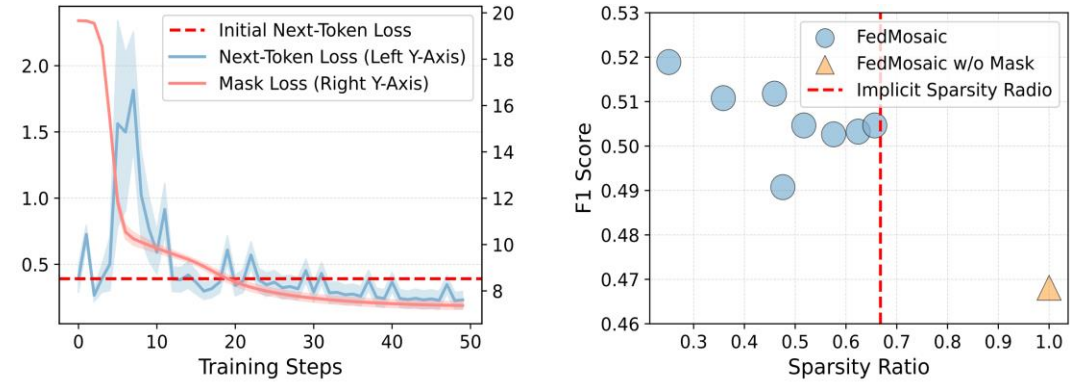


(a) Per-Document Storage

(b) Per-Query Communication

Figure 4: Storage and communication overhead.

Overhead Reduction by Clustering



(a) Training Loss

(b) Mask to Accuracy

Figure 5: Impact of document mask.

Contributions of Document-Specific Mask

Table 5: Accuracy improvement and robustness of selective aggregation under different selection k .

Dataset	Type	w/o Sel.	$k=1$	$k=3$	$k=5$	$k=10$	$k=15$	$k=30$
2WQA	Bridge	0.4224	0.4552 $\uparrow$ 7.77%	0.4447 $\uparrow$ 5.28%	0.4492 $\uparrow$ 6.34%	0.4525 $\uparrow$ 7.13%	0.4525 $\uparrow$ 7.13%	0.4525 $\uparrow$ 7.13%
	Compare	0.5001	0.4940 $\downarrow$ 1.22%	0.5108 $\uparrow$ 2.14%	0.5142 $\uparrow$ 2.82%	0.5142 $\uparrow$ 2.82%	0.5142 $\uparrow$ 2.82%	0.5142 $\uparrow$ 2.82%
	Inf.	0.2137	0.2316 $\uparrow$ 8.38%	0.2581 $\uparrow$ 20.7%	0.2455 $\uparrow$ 14.8%	0.2445 $\uparrow$ 14.4%	0.2422 $\uparrow$ 13.3%	0.2422 $\uparrow$ 13.3%
	Compose	0.0862	0.0750 $\downarrow$ 12.9%	0.0869 $\uparrow$ 0.81%	0.0953 $\uparrow$ 10.5%	0.0953 $\uparrow$ 10.5%	0.0953 $\uparrow$ 10.5%	0.0953 $\uparrow$ 10.5%

Effectiveness of Selective Aggregation

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- We introduce the *first framework for federated RAG* that enforces the *locality constraint*
- We propose *FedMosaic*, a unified design with *Multi-Document Parametric Adapters* and *Selective Adapter Aggregation*
- Extensive evaluations show that FedMosaic achieves *higher accuracy* and *reduces storage and communication cost*

Q & A

THANK YOU

if you have further problems, feel free to email
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